
FNH 200 942

Lesson 03

Fat and Sugar Substitutes
Sensory Perception of Foods

2026 Summer

Mid-term Tuesday, July 21, 2-stage

You need to approach this as a team!!!! Decide what time to start:

6:00 am to 9:30 am (finish by 10) PDT - Stage 1: Individual Exam

- Lockdown Browser required, mock quiz to check technical set up
- Join Zoom (optional), mainly to communicate with me in case anything happen, video off, mic off, volume at low (just in case)
- Canvas, about 30 questions, shuffled, banks, all questions shown
 - Open source, no Proctorio, almost identical settings as the quizzes, but shuffled questions too, with all Qs shown

When everyone in your team finish, join a breakout room, or Discord, or Instagram Group, or Whatsapp Group, or..., start Stage 2 together

- Team, about 20 min; finish by 10:30 am

10:30 to 11:00 Break and Teamwork

11:00 to 11:30 Lesson 05

2023S2

Home

Zoom

Flexible Assessment

Assignments

Discussions

Modules

Piazza

Grades

People

Pages

FNH 200



Flexible Assessment

You have a say on what assessment better reflects your learning

Individual assignments

- Assignment 2 - Individual
- Assignment 3
- Midterms and Final - Individual

Decided by July 21; otherwise, go with default

Learning Objectives

- compare and contrast the different types of fat replacers and alternative sweeteners used in foods.
 - discuss the role that sensory parameters play in our selection and perceptions of food
-

Why fat and sugar replacers?

Do you use them?

If yes, how? What type of foods?

Fat in Our Foods

9 Cal per gram



Fat Replacement

3.1.1 Types of Fat Substitutes

Fat substitutes are grouped as either protein, carbohydrate or fat based. The following

Type of Fat substitute	
Protein based: use protein particles to stabilize and give texture to food. Usually digested	
Simplese ®	Based on soy, milk or egg white protein.

Protein-based - Simplese, Microparticulation, modified proteins

Carbohydrate-based - Maltrins, short chains of starch, maltodextrin

Fat-based - Olean™, a combination of 'sugar' and 'fat'

2. Match the following terms:

(1) Simplesse

(2) Maltrin

(3) Olean

(i) Indigestible

(ii) Partially digestible

(iii) Fully digestible

Protein-based - Simplesse, Microparticulation, modified proteins

Carbohydrate-based - Maltrins, short chains of starch, maltodextrin

Fat-based - Olean™, a combination of 'sugar' and 'fat'

1. Which of the following statement is true for Simplese:

- A. The fat used to make deep-fried products, such as French fries, can be replaced by Simplese.
 - B. Microparticulation is a process used to make tiny spherical Simplese particles.
 - C. Products made with Simplese must be labelled with Simplese in the ingredient list.
 - D. Simplese is a carbohydrate-based fat substitute.
 - E. Simplese is a fat substitute because it is indigestible.
-

1. Which of the following statement is true for Simplese:

- A. The fat used to make deep-fried products, such as French fries, can be replaced by ~~Simplese~~ Olean.
 - B. Microparticulation is a process used to make tiny spherical Simplese particles.
 - C. Products made with Simplese ~~must~~ may be labelled ~~with Simplese~~ as modified protein in the ingredient list.
 - D. Simplese is a ~~carbohydrate~~ protein-based fat substitute.
 - E. Simplese is a fat substitute because it is ~~indigestible~~ partially digestible.
-

Fat Free Pringles



NUTRITION FACTS INFORMACIÓN NUTRICIONAL	
Serving Size 1 Ounce Tamaño de la porción 1 oz	
(28 g, approx./approx. 15 crisps per serving/unidades)	
Servings Per Container approx. 5 Porciones por envase approx. 5	
Amount Per Serving/Cantidad por porción	
Calories/Calorías 70	
Calories from Fat/Calorías de grasa 0	
% Daily Value* % del valor diario*	
Total Fat/Grasa total 0 g	0%
Saturated Fat/Grasa saturada 0 g	0%
Trans Fat/Grasa trans 0 g	0%
Cholesterol/Colesterol 0 mg	0%
Sodium/Sodio 160 mg	7%
Total Carbohydrate	
Carbohidratos totales 15 g	5%
Dietary Fiber/Fibra dietética 1 g	4%
Sugars/Azúcares 1 g	
Protein/Proteína 1 g	
Vitamin A/Vitamina A 0% • Vitamin C/Vitamina C 6%	
Calcium/Calcio 0% • Iron/Hierro 0%	

NUTRITION FACTS INFORMACIÓN NUTRICIONAL

Serving Size 1 Ounce
Tamaño de la porción 1 oz
(28 g, approx./approx. 15 crisps per serving/unidades)
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Calcium/Calcio 0% • Iron/Hierro 0%	

*Percent Daily Values are based on a 2,000 calorie diet. Your daily values may be higher or lower depending on your calorie needs:

*Los porcentajes de valores diarios están basados en una dieta de 2,000 calorías. Sus valores diarios pueden ser mayores o menores, dependiendo de las calorías que usted necesite:

	Calories/Calorías:	2,000	2,500
Total Fat/Grasa total	Less than/Menos de	65 g	80 g
Sat Fat/Grasa saturada	Less than/Menos de	20 g	25 g
Cholesterol/Colesterol	Less than/Menos de	300 mg	300 mg
Sodium/Sodio	Less than/Menos de	2,400 mg	2,400 mg
Total Carbohydrate			
Carbohidratos totales		300 g	375 g
Dietary Fiber/Fibra dietética		25 g	30 g

3. Why is this statement needed on products made with Olestra?

Vitamins A, D, E and K have been added.

- A. Enzymes that digest Olestra may also digest these vitamins.
 - B. Olestra inhibits the absorption of these vitamins.
 - C. Products made with Olestra are usually deep fried; deep frying destroys the availability of these vitamins.
 - D. There is a need to enhance vitamin contents in potato chips.
-

4. Which of the following Olestra containing is commercially available in Canada?

- A. Low fat yoghurt
 - B. Processed meats
 - C. Potato chips
 - D. All of the above
 - E. None of the above
-

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Lesson 03

Fat and Sugar Substitutes
Sensory Perception of Foods



2026 Summer

Sweetness Index

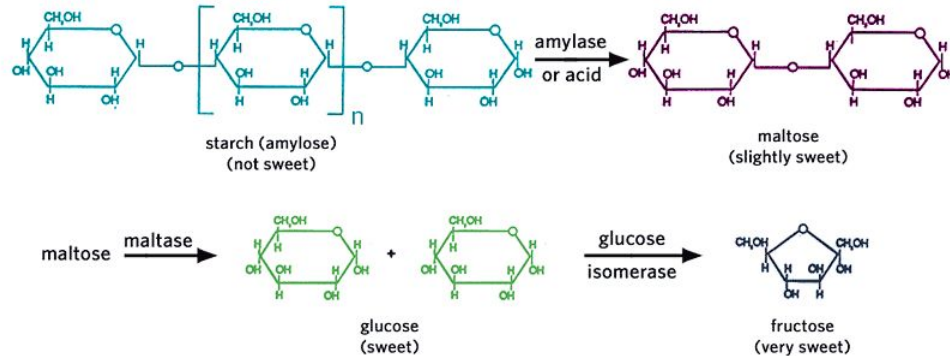


Table 2.2. Relative sweetness of carbohydrate

Sugar	Sweetness Index*
sucrose	100
glucose (dextrose)	70 - 80
fructose (levulose)	140
invert sugar	100 - 130
corn syrup (mixture of glucose, maltose)	50
maltose	20
lactose	10 - 20
galactose	60
sorbitol	50
xylitol	100
high fructose corn syrups:	
42% fructose	100
55% fructose	100+
90% fructose	120 - 160

Sweetness Index

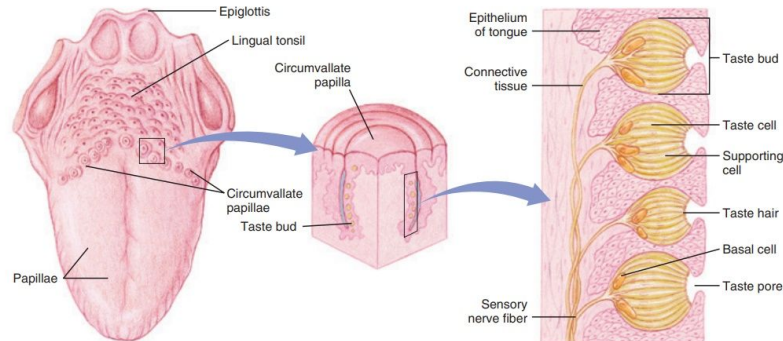
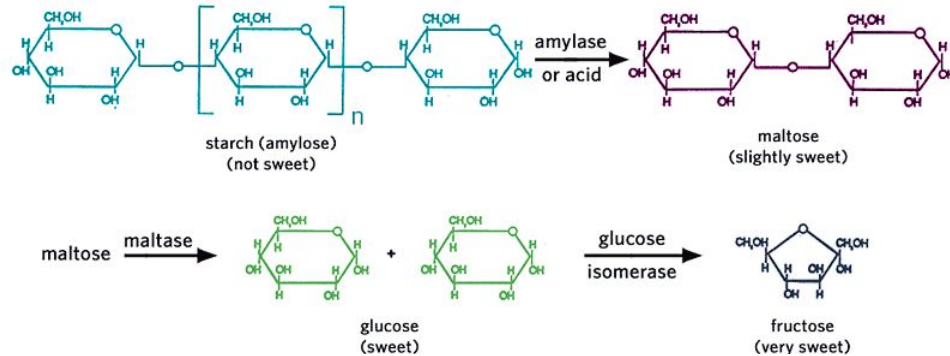


FIGURE 15-2 The taste buds of the tongue.

Table 2.2. Relative sweetness of carbohydrate

Sugar	Sweetness Index*
sucrose	100
glucose (dextrose)	70 - 80
fructose (levulose)	140
invert sugar	100 - 130
corn syrup (mixture of glucose, maltose)	50
maltose	20
lactose	10 - 20
galactose	60
sorbitol	50
xylitol	100
high fructose corn syrups:	
42% fructose	100
55% fructose	100+
90% fructose	120 - 160

Sweet...

Sweetening agents:

Food and Drug Regulation - Division 18

Honey, sugar, molasses

Sweeteners:

Food and Drug Regulation - Division 16

Food additives

Food and Drug Regulations - Lesson 04



Search

Family Law	Criminal Justice	Funding	Canada's System of Justice	Laws
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[Home](#) → [Laws Website Home](#) → [Consolidated Regulations](#) → [C.R.C., c. 870 - Table of Contents](#) → [C.R.C., c. 870](#)

Constitutional Documents

Canadian Charter of Rights and Freedoms

Consolidation of Constitution Acts, 1867 to 1982

Consolidation of Constitution Acts, 1867 to 1982 (1990 Report version)

Food and Drug Regulations (C.R.C., c. 870)

Full Document: [HTML](#) (Accessibility Buttons available) | [XML](#) [4724 KB] | [PDF](#) [12995 KB]

Regulations are current to 2023-06-21 and [last amended](#) on 2023-02-15. [Previous Versions](#)

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PART B

Foods (continued)

DIVISION 17

Salt

Zero/Low Calories?

What other low-calorie products can you identified?



A. Aspartame

B. Cyclamate

C. Sorbitol

D. Sucralose

E. Acesulfame-K

A. Aspartame

B. Cyclamate

C. Sorbitol (sugar alcohol)

D. Sucralose

E. Acesulfame-K



Two types:

Digestible - Low Caloric

Indigestible - Non Caloric

A. Aspartame (Low), two amino acids, 200x, PKU

B. Cyclamate (Non)

C. Sorbitol (Low), sugar alcohol, less sweet, cooling

D. Sucralose (Non), 600x, heat stable

E. Acesulfame-K (Non), 200x, heat stable

5. Which sugar substitute provides a cool-refreshing sensation in chewing gum?

A. Aspartame

B. Cyclamate

C. Sorbitol

D. Sucralose

E. Acesulfame-K

Stevia



Monk Fruit



TABLE IX

Food Additives That May Be Used as Sweeteners

Item	Column I	Column II	Column III
No.	Additive	Permitted in or on	Maximum Level of Use
A.01	Acesulfame-potassium	(1) Table-top sweeteners	(1) Good Manufacturing Practice
		(2) Carbonated beverages	(2) 0.025% in beverages as consumed
		(3) Unstandardized beverage concentrates; Unstandardized beverage mixes; Unstandardized beverages; Unstandardized dairy beverages	(3) 0.05% in beverages as consumed
		(4) Filling mixes; Fillings; Topping mixes; Toppings; Unstandardized dessert mixes; Unstandardized desserts; Yogurt	(4) 0.1% in products as consumed
		(5) Chewing gum; Breath freshener products	(5) 0.35%
		(6) Unstandardized fruit spreads	(6) 0.1%

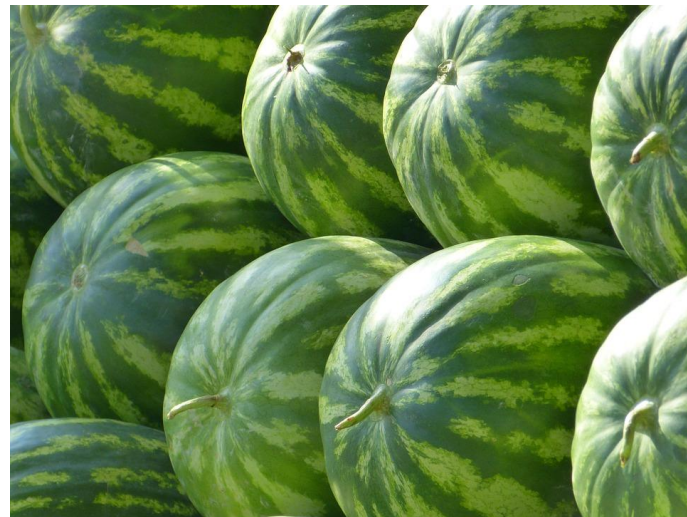
Sensory Properties of Foods

What makes a food good (yummy)?

How Do You Pick the Best Apple?



How Do You Pick the Best Watermelon?



6. How do we describe the force needed to squeeze a loaf of bread?

- A. Compression
 - B. Cutting
 - C. Shearing
 - D. Tensile strength
-

7. How do we perceive aroma?

- A. Fat soluble volatiles interacting with receptors in our nasal passages
 - B. Fat soluble volatiles interacting with receptors on our tongue
 - C. Water soluble volatiles interacting with receptors in our nasal passages
 - D. Water soluble volatiles interacting with receptors on our tongue
-

Taste

What are the four tastes?

Is there a fifth? How about MSG?

MSG

Monosodium
glutamate

Favour enhancer

- Savory
- Suppress bitterness



8. Match the oral sensations with the responsible molecules.

(i) Astringency

(ii) Pungency

(iii) Coolness

(a) Xylitol

(b) Capsaicin

(c) Tannins

A. (i)-(a), (ii)-(b), (iii)-(c)

B. (i)-(a), (ii)-(c), (iii)-(b)

C. (i)-(b), (ii)-(a), (iii)-(c)

D. (i)-(b), (ii)-(c), (iii)-(a)

E. (i)-(c), (ii)-(b), (iii)-(a)



Three Common Tests

Hedonic Test - Consumer

Discriminative Test - Products (aka Difference Test)

Descriptive Test - Products, details

9. Which analysis offers the most details during product development?

- A. Discriminative Analysis
 - B. Difference Test
 - C. Descriptive Analysis
 - D. Hedonic Test
-

10. This test is best used to measure consumer acceptance level of a new product.

- A. Discriminative Analysis
 - B. Difference Test
 - C. Descriptive Analysis
 - D. Hedonic Test
-

Summary

- Fat substitutes can be made from proteins, carbohydrates or even some types/forms of fats! Different fat substitutes provide food with unique characteristics
 - Sugar substitutes are also referred to by the public as "Artificial Sweeteners"; vary in composition, sweetness level, calories, and applications.
 - Sensory factors: Appearance, texture and flavour (includes aroma and taste)
 - Sensory Evaluation of foods is a scientific discipline widely used to measure and quantify our perceptions to food.
-

How are your teammates?

What are some strengths you can contribute to your team?

Teamwork - Get into Teams

Introduce yourself, video on, mic on

Team Contract (1 out of the 20%):

- Please download a copy
- Copy and paste text and submit it on Canvas

Team Assignment (5% out of 10%):

- Start brainstorming idea today; sign-up your topic on UBC Wiki
 - Do some research this weekend
-

Assessments

- Assignment #1 - Thanks. I am enjoying your posts and questions
 - Teams - should be quite set by now
 - Team contract & Assignment #2 Team portion
 - Submit what you have, if your MIA teammate shows up later, the MIA teammate can re-submit it later
-

Piazza

- I'm monitoring...
 - Please help each other. You are often better at explaining something to your peer (right tone, right depth)
 - Anonymous... if someone makes a mistake, please correct... I will certainly intervene at times.
-

Team Project

- A. An information interview
- B. A 'traditional' research paper

Mid-term Tuesday, July 21, 2-stage

You need to approach this as a team!!!! Decide what time to start:

6:00 am to 10:00 am PDT - Stage 1: Individual Exam

- Lockdown Browser
- Join Zoom (optional), mainly to communicate with me in case anything happen, video off, mic off, volume at low
- Canvas, about 30 questions, shuffled, banks, all questions shown, 40 mins
 - Open source, no Proctorio, almost identical settings as the quizzes, just shuffled questions, with all Qs shown
- When everyone in team finish, start Stage 2: Team, about 20 min
- Both stages done by 10:30 am

10:30 to 11:00 Break and Teamwork

11:00 to 11:50 Lesson 05
